

1 Robustness — what & why

Robustness = ability of AI to keep **optimal performance & reliability** under uncertainty / adversarial conditions; generalize well & give reliable predictions despite **noisy data, distribution shifts, unforeseen extreme events, adversarial attackers**. Bridges from *AI Risks* to robustness via risk decomposition.

2 Black Swans, Long Tails, Unknown Unknowns

Black Swans = outlier events outside typical expectations, carrying **extreme impact** (name: Europeans assumed all swans white until black ones found in Australia). Costly to ignore — they matter most. Ex: 2008 crisis, COVID-19.

Long-tail distribution: tail = region far from the head/centre; tapers off **gradually** (Pareto / 80-20). **Power laws are 'scale-free'** (e.g. Web degree distribution). Long tails are pervasive: ~0.1% of drugs ≈ 50% pharma sales; ~1% of artists ≈ 77% music revenue.

Mediocristan vs Extremistan

Mediocristan	Extremistan
Thin tails	Long tails
Total from many small events	Total from a few large events
Typical member = average	Member giant or dwarf
Tyranny of the collective	Tyranny of the accidental; top few dominate
Easy to predict; mild	Hard to predict; wild; scalable impact

Nonlinear interactions generate long tails — result is long-tailed when variables are **multiplied** (would be thin-tailed Gaussian if added). Occurs when parts are interdependent / output → 0 if a part → 0. **Research output = Ideas × Time × Students × Resources**.

Black Swans dominate risk analysis → we use long tails to characterise them statistically. Other x-risks (biorisks, asteroids) are extreme long-tail events. Future AI impact may be long-tailed → want models that **withstand & detect Black Swans**.

Unknown Unknowns (Rumsfeld matrix)

Known	Unknown
Known-Knowns : aware+understand; facts, recollection	Unknown-Knowns : understand but unaware; tacit knowledge, self-analysis
Known-Unknowns : aware but don't understand; classic risks, conscious ignorance, closed-ended Qs	Unknown-Unknowns : neither aware nor understand; meta-ignorance, open-ended exploration

Assignment: Black Swans = outliers with extreme, unpredictable impact. 'Known-Unknowns' = **Closed-ended Questions + Conscious Ignorance**. Long-tail equation = things **multiplied** (Idea × Student × Resources × Time). Extremistan = tyranny of the accidental + scalable impact; Mediocristan = tyranny of the collective.

3 Measuring & Improving Long-Tail Robustness

Stress-test datasets simulate extreme/unusual events from a **different data-generating process** than training. Goal: performance shouldn't **degrade sharply** under stressors.

ImageNet stress tests:

- **ImageNet-C** (Corruptions) — blur, noise, weather; ResNet-50 gets 76% on ImageNet but drops sharply
- **ImageNet-R** (Rendition) — 30K art/cartoon/graffiti/origami styles; tests style robustness; disjoint from ImageNet
- **ImageNet-A** (Adversarial/natural) — real examples that fool ResNet-50; also fool ViTs → **shared weaknesses across architectures**
- **ObjectNet** — new viewpoints & backgrounds
- **ANLI** — adversarial NLI built by crowdworkers to fool models; GPT-3 only ~40%

Augmentation methods (improve robustness):

- **Larger models** generalize better (more redundancy: one neuron fails, another detects the feature)
 - **Mixup** — elementwise convex combination of inputs+outputs
 - **AutoAugment** — searches augmentation strategies (Invert, Solarize...); trains 10Ks of nets → costly
 - **AugMix** — random augmentations mixed to keep images recognizable
 - **PixMix** — mixes training images with a **separate** set (e.g. fractals)
- Note: many methods improve **safety metrics but not typical accuracy**. Natural adversarial examples = mine hard/high-confidence-wrong examples.

4 Distribution Shifts

Occurs when the **joint distribution of inputs & outputs differs between train and test**. Present in nearly every practical ML app (can't perfectly reproduce test conditions at train time).

Type	What changes	Example
Covariate shift	P(x) changes, P(y x) stays . Model fits train well, fails on test	Driverless car: sunny train → snow test; native-English speech → all-English test
Concept shift	P(x) same, but relation / decision boundary P(y x) changes	Pre-pandemic browsing history used March 2020 for recommendations
Prior-probability / label shift	Reverse of covariate . Only in $y \rightarrow x$ problems: P(y) changes, P(x y) stays	Flu season: test(flu) > train(flu) but symptoms flu unchanged

Detecting shift: (1) monitor the **model** — accuracy/precision/metrics drifting over time; (2) monitor the **data** — compare statistical properties of training vs deployment data.

5 Data Poisoning — Trojan Attacks (attacker has data access)

Trojan / backdoor = implant **hidden functionality** by poisoning the dataset. Model behaves normally until a secret trigger appears, then misbehaves.

- **Neural Cleanse** — defense: detects & reverse-engineers backdoor triggers
- **Nightshade** — 'offensive' poisoning tool: corrupts training images to protect artists' work from scraping

6 Adversarial Robustness (Attacks & Defenses)

Adversarial examples = inputs perturbed to fool a model. Modern models are robust to **imperceptible** distortions but not **perceptible** ones (e.g. stickers make a Stop sign read as 'Speed 45' — 100% success in lab, 85% in field).

Attacks:

- **FGSM** (Fast Gradient Sign Method) — **one** gradient-ascent step to maximize loss within a p-norm budget ϵ ; fast but **easy to defend**
- **PGD** (Projected Gradient Descent) — **iterative** FGSM, projects back into ϵ -ball; stronger
- **Targeted** — minimize target-label loss + maximize true-label loss (better when many similar labels)
- **Text attacks** — insert/delete/replace at char / word / sentence / multi level (TextAttack library)
- **Jailbreaks** — e.g. DAN; universal & transferable prompt attacks (llm-attacks.org)

Access models:

- **White-box** — full access (params, architecture)
- **Black-box** — no access, only query outputs; more realistic
- **Transferability** — examples crafted for one model fool **other** models (ResNet50 attack works on other CNNs)

Defenses:

- **More data** — helps (adversarial pretraining on ImageNet) but data is a bottleneck
- **Data augmentation**
- **Adversarial training** — train on adversarial examples; strongest, but **trade-off: lowers clean accuracy**

Assignment: Covariate shift = P(x) changes, P(y|x) same. Transferability = attacks for one model fool another. Adversarial-training trade-off = **reduces accuracy on clean examples**.

7 RLHF — Reinforcement Learning from Human Feedback

Goal: optimize a language model for **human preferences** (e.g. summarization). For instruction x and sample y we want a human reward $R(x,y) \in \mathbb{R}$, higher = better, and maximize expected reward.

Two problems & fixes:

- **P1: humans-in-the-loop are expensive** → train a separate **Reward Model** $RM_\phi(x,y)$ to predict human reward from an annotated dataset, then optimize against RM_ϕ
- **P2: human ratings are noisy & miscalibrated** → ask for **pairwise comparisons** ($y_w > y_l$), more reliable; modelled by **Bradley-Terry (1952)** — winner should score higher than loser

Scaling & products:

- **InstructGPT** — scales RLHF to ~50K tasks; prompt kinds: plain, few-shot, user-based
- **ChatGPT** = instruction finetuning + RLHF for dialog
- RLHF > pretraining+finetuning when tuned well; but computationally expensive & tricky (value function, slow online sampling, hyperparameter-sensitive)

8 AI Alignment & RLHF Challenges

Alignment problem: can the system act with **human intentions & values**? Machines solve problems fast but may not listen to instructions; must be **incentivized to tell the truth** (hallucination). Hard to tell AI- vs human-generated text.

RLHF ablations (exam):

- RLHF – Human Feedback = **RLAIF** (RL from AI Feedback)
- RLHF – Reward Model = **DPO** (Direct Preference Optimization)
- RLHF – RL = **Supervised Finetuning (SFT)**

State of alignment: RLHF was *never* designed to solve alignment, but is the current state-of-the-art for it.

The RLHF pipeline (3 steps):

- **1. Instruction tuning / SFT** — collect **demonstration** data, train a supervised policy (*this is step 1*)
- **2. Reward model** — collect comparison data, train RM_ϕ
- **3. RL optimization** — generate completions, score with RM_ϕ , update policy to raise probability of high-reward completions

Add a **KL-divergence penalty** so the policy doesn't **drift too far from the pretrained model** (rewards are imperfect → avoid over-optimization).

Reward hacking / proxy gaming:

- Optimizing an **imperfect proxy** → unintended behaviour. Classic: boat-race RL agent grabs turbo-boosts & loops for points instead of finishing the race
- Real world: YouTube/Insta optimize **engagement** → harms mental health
- We risk **losing control** as AIs get more capable

Three families of RLHF challenges:

Human feedback: hard to pick representative humans; evaluator biases; individuals can poison data; simple mistakes; partial observability; can be misled/gamed; richness-vs-efficiency trade-off.

Reward model: one person's values hard to encode; a single reward can't represent a diverse society; imperfect proxy → **reward hacking**; costly to evaluate.

Policy: poor in adversarial situations; poor in deployment even with perfect rewards; optimal RL agents **seek power**; RL causes **mode collapse**.

Fix: transparency & **auditing** — but **black-box access is insufficient** (auditors can only query & observe outputs, not inspect internals).

9 Tutorial — Transformers & Hugging Face (notebooks)

- **Lineage:** RNN → Seq2Seq → Attention → **Transformer** ('Attention is all you need')
- **Self-attention + Multi-Head Attention:** Encoder (feed-forward, embedding + **positional** encoding, masks) & Decoder
- **Revolutionary:** compute-efficient, embodies **the Bitter Lesson**, generalizes across domains
- **Limits:** high memory, large compute/data, token-length limits. Leads to **BERT & GPT**
- **Tokenization:** per-char (small vocab, long seq) vs per-word (huge vocab); LLMs use **subword** — **Byte-Pair Encoding**, **WordPiece**, **SentencePiece** (learn frequent substrings from a corpus)
- **Hugging Face:** tokenizers, inference pipelines, custom training loop, fine-tuning with the **Trainer API**